

# AfterDark Jukebox



*A token-operated video jukebox running on a Raspberry Pi 5. An IR beam sensor detects when a token is inserted, the user selects a video by pressing a letter then a number on a keyboard, and the video plays full screen. An LED strip dims and brightens in sync with the state of the jukebox. If another token is inserted before the video finishes the current content will fade and wait for another selection to be made.*

*The app runs on a single small screen. A second, larger screen can stay connected for development without the app appearing on it.*

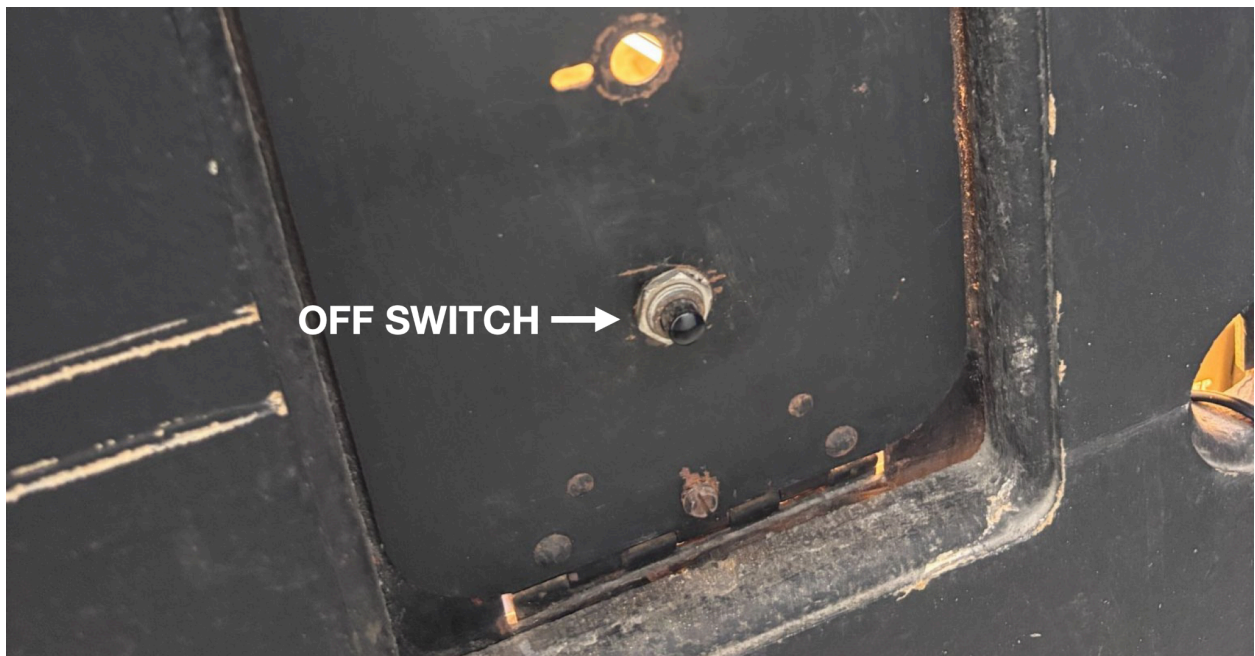
## 1. Turning On

To turn on, simply switch on at the power. After about 60 seconds the screen should turn on, and the jukebox software will load automatically. If any warning messages appear, they should disappear on their own after about 10 seconds.

If the jukebox appears to be off when you want it on, just unplug it and plug it back in again. This will restart the booting process.

## 2. Turning Off

The safest way to turn off is to hold down the off button located on the back for **5 seconds**. This powers down the computer inside safely.



**Please don't just cut the power at the socket** — doing this instead of using the button risks corrupting the memory card inside, which would need a replacement card or a full rebuild to fix.

## 3. Troubleshooting

### Jammed token

If a token gets jammed, this will happen right at the top of the coin chute. Try poking it gently with something if you can reach it, or try popping another token in.

### Screen does not turn on

Make sure the HDMI cable is plugged into both the screen and the Raspberry Pi. Also check the USB-C power is plugged into the screen, and the Raspberry Pi's own power cable is connected too.

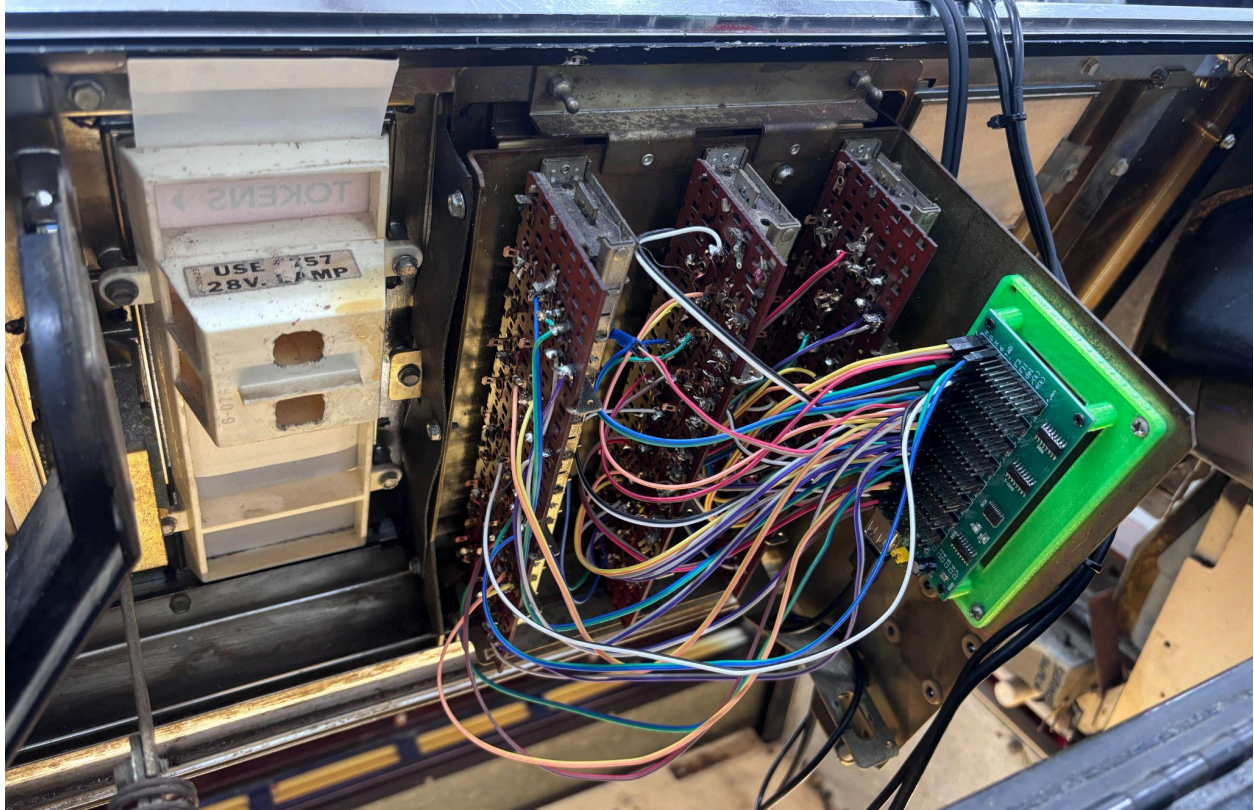
### The Raspberry Pi turns on but the screen shows error messages

This normally means the SD card in the Raspberry Pi (its memory) has become corrupted. The best fix is to open the Raspberry Pi box and replace the SD card with the spare provided. If this doesn't work, a full rebuild may be needed — instructions for this are kept here:

[https://github.com/michaelshorter/AfterDark\\_Final](https://github.com/michaelshorter/AfterDark_Final)

## No selections can be made

Make sure the USB cable from the buttons is connected into the Raspberry Pi.



## 4. Updating the Content

To update the content, remove the USB stick (labelled **JUKEBOX**) and add your files to it. On there, you'll find:

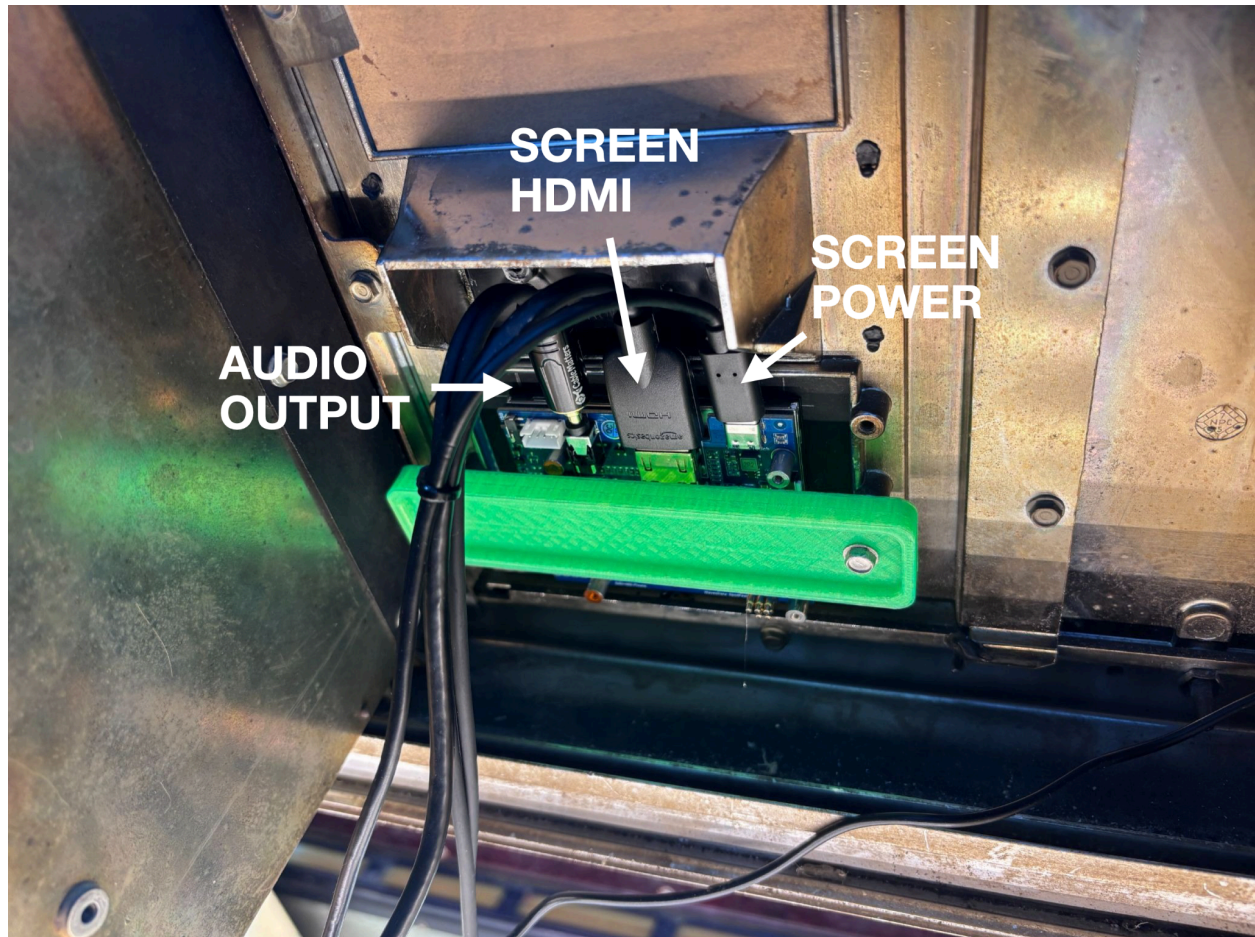
- Two images: insert\_token.png and make\_selection.png — these are exactly what they say they are, shown on screen while waiting for a token and while waiting for a selection.
- A folder called videos, containing files named by letter and number, e.g. A1.mp4, A2.mp4, B3.mp4. When a selection is made, the jukebox plays the file matching that letter and number exactly.

A few things worth knowing when adding new videos:

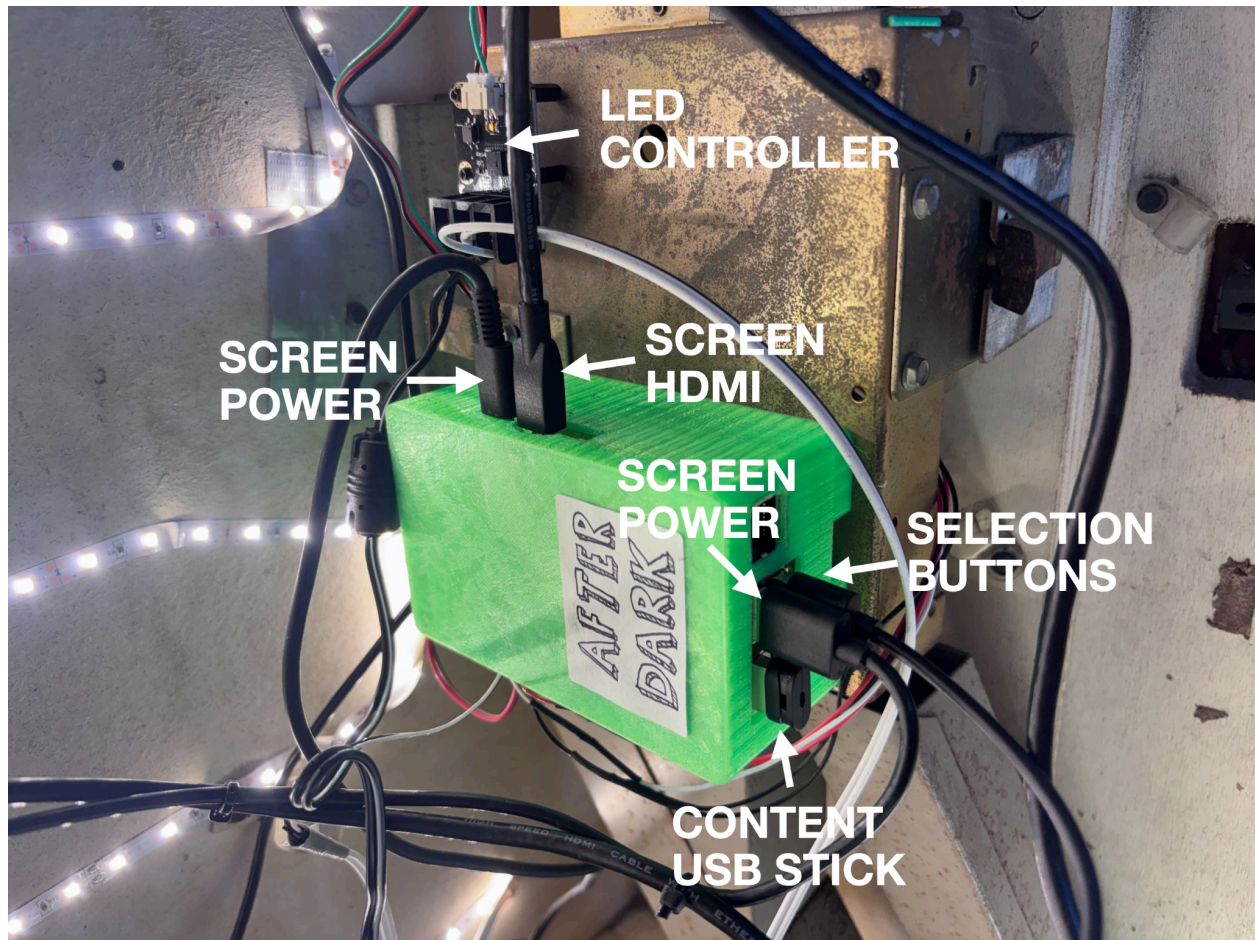
- Valid letters are A–H, J–N, Q–V (the letters I, O, and P are skipped, to avoid confusion with the numbers 1 and 0 on screen). Valid numbers are 0–9.
- Every video file must sit inside the videos folder, not loose on the USB stick itself.
- File names must match the letter/number exactly (capital letter, no spaces), or the jukebox won't find them.

## 5. If the Screen Goes Squint

The screen is clamped in place with the green bracket. You should be able to move it gently back into place. You can also loosen this bracket if it needs moved more — but be careful here, as loosening it may let the screen fall out along with its paper frame, and fragile glass.



Wiring Setup



**Parts List:**

| Part                        | Link                                                                            |
|-----------------------------|---------------------------------------------------------------------------------|
| Raspberry Pi 5              | <a href="#">Raspberry Pi 5</a>                                                  |
| Raspberry Pi 5 power supply | <a href="#">Raspberry Pi 27W USB-C Power Supply</a>                             |
| USB keyboard module         | <a href="#">CH9328 USB HID Keyboard Module</a>                                  |
| IR sensor                   | <a href="#">IR Break Beam Sensor - 3mm LEDs</a>                                 |
| Small screen                | <a href="#">3.5" HDMI Capacitive Touch IPS LCD Display (640×480)</a>            |
| LED strip                   | <a href="#">Double-Sided Single Color 12V LED Strip - Warm White 3000K - 1m</a> |
| LED controller              | <a href="#">Gravity: MOSFET Power Controller</a>                                |
| LED power supply            | <a href="#">Mean Well 5V 4A 20W Power Supply</a>                                |
| HDMI → micro HDMI cable     | <a href="#">Raspberry Pi Micro-HDMI Cable</a>                                   |
| USB to USB-C cable          | <a href="#">USB-A to USB-C Cable</a>                                            |
| USB stick                   | <a href="#">Raspberry Pi Flash Drive</a>                                        |